PredGAN - a deep multi-scale video prediction framework for detecting anomalies in videos

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ABSTRACT

In this paper we propose a multi-scale video prediction framework with adversarial training for detecting anomalies in videos. Anomalous events are those which do not conform to normal behavior. Supervised learning framework cannot account for all the unusual activities since a universal definition of anomaly cannot be adopted. To tackle this problem, we propose an unsupervised approach to learn the internal representation of videos and use this learning to accurately predict the future-frames of the videos. We train our network adversarially on videos consisting of only normal activities. When our network encounters unusual or irregular activities the generated frames consists of fuzzy regions where the irregular activities are present. These fuzzy regions consequently lower the peak signal to noise ratio (PSNR) of the generated frames. The PSNR values are normalized to have values between 0 and 1 and is used as a regularity score to tag a frame as anomalous or not-anomalous. We provide quantitative and qualitative evaluation of the proposed framework and also introduce Earth Mover's Distance as a new evaluation metric to assess the quality of the images generated. We demonstrate our framework on UCSD Pedestrian dataset and show that we achieve comparable results.

CCS CONCEPTS

• **Computing methodologies → Scene anomaly detection**.

KEYWORDS

Anomaly detection, video frame prediction, generative adversarial networks

ACM Reference Format:
Adarsh Jamadandi, Sunidhi Kottursheettar, and Uma Mudenagudi. 2018. PredGAN - a deep multi-scale video prediction framework for detecting anomalies in videos. In *11th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP 2018), December 18–22, 2018, Hyderabad, India,* Anoop M. Namboodiri, Vineeth Balasubramanian, Amit Roy-Chowdhury, and Guido Gerig (Eds.). ACM, New York, NY, USA, Article 001, 8 pages. https://doi.org/10.1145/3293353.3293354

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ICVGIP 2018, December 18–22, 2018, Hyderabad, India
© 2018 Association for Computing Machinery.
ACM ISBN 978-1-4503-6615-1/18/12.
https://doi.org/10.1145/3293353.3293354

1 INTRODUCTION

In this paper we address the problem of anomaly detection in surveillance videos. Anomalous events are those which do not conform to normal behaviour and detecting such unusual activities in videos is an important application in surveillance domain. However, detecting such activities in videos forms a non-trivial task as the definition of anomaly is subjective and real-world scenarios exhibit a wide array of situations which are impossible to account for using a classification method. We propose an adversarial training procedure to learn the internal representation of videos consisting of normal events and use this learning to accurately predict the evolution of the video frames. Generative Adversarial Networks (GAN) introduced by Goodfellow et. al [5] has become a popular candidate for image generation tasks. GAN has proved to be an apt framework for various applications like image inpainting [2, 20], image super-resolution[8] and also in various 3D applications like shape completion and inpainting of 3d models [19]. GANs consist of two networks, a generator whose job is to generate fake data samples which look realistic, it is modeled to transform a random vector z to an image I. The random vector z is usually sampled from a uniform distribution and is expected to generate images I which are close to the training data (P_{data}) and the job of a discriminator which is usually a binary classifier is to output a lower probability for data which are being generated by the generator. The two networks are trained adversarially and the training process ends when the discriminator is unable to distinguish "real" and "fake" data.

Most of the anomaly detection methods in the literature [1] talk about using either hand-crafted features to learn a dictionary of normal events and then try to parametrize the reconstruction error, which requires an ensemble of different hand-crafted features to accommodate all kinds of anomalies and the dictionary learnt could prove to be over-complete or the latest trend involves using deep learning techniques like convolutional neural networks (CNN) or auto-encoders to use the reconstruction errors to detect anomalies, the problem with this approach is that deep learning techniques can [6, 13] effectively learn features thus leading to smaller reconstruction errors. To tackle these problems, we propose an anomaly detection framework that compares the generated frame with the ground truth to detect anomalies, specifically we use a convolutional neural network which is trained in an adversarial manner as proposed by Mathieu et. al in [12] to learn the internal representation of videos consisting of only normal events, the network is able to predict the future frames of a given video clip accurately

which falls under the realm of training data, only the last few clips are either generated by the network or real, the remaining frames are usually from the training data, this helps the network to make use of the temporal information. When a video clip which contains irregular activities is fed to the network, the network fails to accurately predict the frames consisting of the irregular activities leading to a blurry or a fuzzy region in the frame prediction where the irregular activity is encountered. We compare the generated frame and the ground truth to parametrize it as a regularity score to decide whether a given frame is anomalous or not. We also introduce a new metric called Earth Mover's Distance (EMD) as one of the evaluation criteria to decide whether the frames are anomalous or not-anomalous. To the best of our knowledge, this is the first work that leverages EMD as a criteria to evaluate the results of video prediction framework. We summarize our contribution as follows :

(1) We propose a video prediction framework to detect anomalies in videos. The network is trained on normal events and is able to accurately predict the evolution of the video frames, the frames consisting of abnormal events are predicted fuzzily which consequently lower the PSNR value, thus allowing us to detect anomalies in videos.
(2) We propose a new evaluation criteria based on Earth Mover's Distance (EMD) to assess the quality of the frames generated and to decide whether a frame is anomalous. We present a quantitative explanation comparing the performance of PSNR and EMD on generated frames, and show that EMD is a more suitable candidate to assess the quality of the images generated.
(3) We demonstrate our results on the popular anomaly detection datasets like UCSD pedestrian dataset 1 and 2 and show that we achieve comparable results to existing techniques.

2 RELATED WORK

Anomaly detection can be treated as a binary classification problem provided the ground truth labels are available. However, this approach proves infeasible because of lack of annotated datasets and the inability to account for all the scenarios with hand-labeled ground truths. Literature [1] talks about using hand-crafted features to model the video data, the problem with this approach is that the features used might not be ubiquitous and do not scale well with changing scenarios. Deep learning techniques have been extremely successful in variety of computer vision tasks and convolutional neural networks (CNN) have had success in modeling image data, for example, Hasan et. al in paper [6] proposed a CNN based anomaly detection framework but the problem is CNNs do not consider the temporal dependencies that are necessary to model video data. Recurrent Neural Networks (RNN) which were mainly introduced to tackle the problems concerning sequence prediction and modeling are being extensively used to model video data because RNNs consist of memory blocks called Long Short-Term Memory (LSTMs) which can account for the long-term dependencies that exist in the data. LSTMs have been explicitly used for image caption generation [7], describing visual content using natural language text [18].

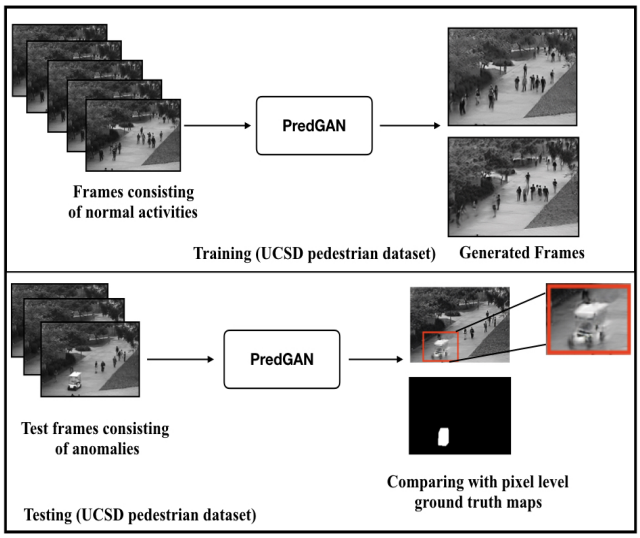

Figure 1: This is the pipeline of our framework. We propose a video prediction framework which we refer to as PredGAN that is trained adversarially and conditioned on normal events. We compare the generated frames with the pixel-level ground truth image to detect anomalies.

Medel et. al [13] uses a composite structure called Convolutional-LSTM (Conv-LSTM) to model the video data and use the sequential nature of LSTMs to predict future video frames, the problem with this approach is that the predicted frames are blurry and lack sharpness. To alleviate this problem we propose a multi-scale video prediction framework which is trained adversarially to predict the evolution of the future frames by taking a sequence of frames as input, the results generated are visually pleasing and the images retain their sharpness.

3 PROPOSED METHODOLOGY

In this section we discuss the proposed methodology and the network architecture details. Figure 1 shows an overview of our proposed approach.

3.1 Video Prediction Framework

Generative Adversarial Networks (GANs) proposed by Goodfellow et. al [5] has garnered a lot of interest from the research community, it has been applied to variety of image generation tasks like image inpainting [20], 3D shape completion [19] etc. A GAN network consists of two networks, a generator $\mathcal{G}$ and a discriminator $\mathcal{D}$ competing against each other in a zero-sum game[14], the generator's job is to sample from a latent space provided by the training data and generate candidates which are realistic, the discriminator which is usually a binary classifier is involved in discriminating whether the samples are real or generated by the generator. We use this approach to train a convolutional neural network that takes a series of video frames as input and predicts the evolution of the

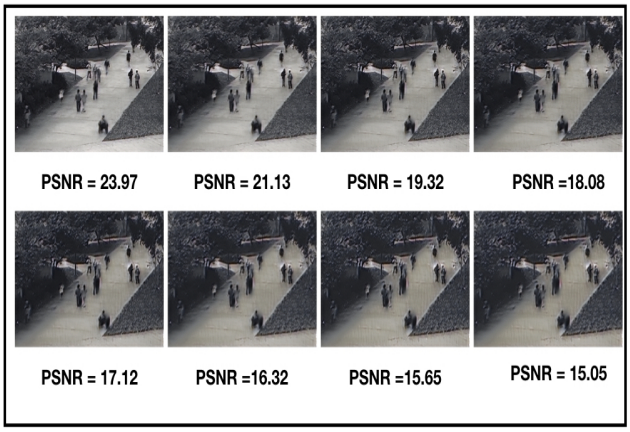

Figure 2: To obtain sharper image predictions we set the number of frames generated equal to the number of input frames. In this case we have experimentally set the input frames to be 5. It can be seen in the figure that, as the number of frames generated increases, the PSNR value decreases.

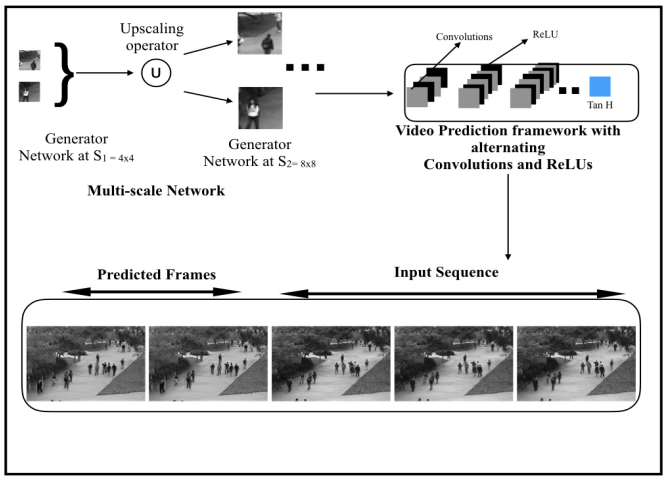

Figure 3: We use a video prediction framework based on alternating convolutions and ReLUs and use a multiscale network to prevent loss of resolution from the convolutions performed. The multiscales can be linearly combined using a Laplacian pyramid.

future frames. Our network architecture is similar to the one as proposed by Mathieu et. al in [12], the discriminator takes a sequence of video frames and predicts the probability of whether the frames are generated by the generator or are from the training data, note that only the last frames are either generated or are from training data, the rest of the frames are the input data. This means that setting the number of generated frames less than or equal to the number of input frames allowed the network to generate sharper video frames with higher PSNR values. In our experiments, we set the input frames to be 5 and allowed the network to generate next 5 frames. This is shown in Figure 2. where the input frames I is set equal to 5. The PSNR values indicate the decrease in the quality of the frames generated. The feeding of sequence of input frames also allows $\mathcal{D}$ to make use of the temporal information, allowing $\mathcal{G}$ to generate frames which are temporally coherent unlike the methodology proposed in [10] which enforces an optical flow constraint to account for the temporal consistency.

3.2 Multi-scale network

In literature [12], the authors discuss about the problem of accounting long-range dependencies of the input data and preserving the resolution of the input data when using a convolutional neural network that consists of alternating convolutional and Rectified Linear Units (ReLU). To go about this, the authors have proposed a multi-scale network which accepts input data at various scaled versions and then later they are combined linearly in a Laplacian Pyramid model [3]. The frame prediction model works by trying to minimize the l_p distance (where p = 1,2) between the true frame and the predicted frame. Mathematically, the l_p distance is given by,

$$\mathcal{L}_p(M, N) = \|G(M) - N\|_p^p \tag{1}$$

We use this approach to build a convolutional neural network that accepts the video frames at various scales of resolution of the input frames. Let $s_1, s_2, s_3, ...s_X$ be the sizes of the input image $\mathcal{I}$, the network starts with the lowest available resolution of $s_1 = 4x4$ and uses this scaled value to proceed to the next resolution. We proceed in this manner till we reach a patch size of $32x32$, it is possible to incorporate sizes greater than $32x32$, but since the discriminator has fully-connected layers after the convolutional layers and an image patch size of $256x256$ passing through 128 feature maps and a fully connected layer of 512 nodes would require weights equal to 429,49,67,296. Thus, higher resolution implies we would need a greater GPU capacity, note that all the training is done on a local machine with 8GB RAM and an i5 processor with no dedicated GPU support. Since the network is fully convolutional the test images can be of higher resolution. The motivation for including a multi-scale network is to reduce the memory restrictions, thus enabling us to incorporate this framework in real-time applications. The network architecture is shown in figure 2.

3.3 Loss functions

In this section we discuss the loss functions imposed on the generator and the discriminator network. Anomaly detection refers to picking out unusual activities, for this we use a next frame prediction network proposed in [12] to learn the normal activities of the video frames. That is, the generator $\mathcal{G}$ is conditioned on normal activities allowing it to generate video frames which are more similar to the activities present in the normal frames.

(1) **Training the discriminator ($\mathcal{D}$)** : The discriminator network is a binary classifier that has to classify the samples into two classes 1 or 0, that is, real or generated data. The training of $\mathcal{D}$ is carried out by fixing the weights of $\mathcal{G}$. The loss function imposed on $\mathcal{D}$ is similar to the one proposed by Mathieu. et al in [12] unlike the mean square error loss function proposed in [10]. For a set of datapoint (M,N) at

each scale s_x (as discussed in section 3.2) a stochastic gradient descent (SGD) is carried out with the loss function given by,

$$\mathcal{L}_{\mathcal{D}}(M,N) = \sum_{k=1}^{Xscales} L_{bce}(D_k(M_k, N_k), 1) + L_{bce}(D_k(M_k, G_k(M)), 0)$$

where, L_{bce} is binary cross entropy loss given by,

$$L_{bce}(Y, \hat{Y}) = -\sum_i \hat{Y}_i log(Y_i) + (1-\hat{Y}) log(1-\hat{Y}) \quad (2)$$

(2) **Training the Generator ($\mathcal{G}$) :** The generator network which is conditioned on the normal activities will try to produce samples which are difficult for the discriminator to distinguish, in an ideal scenario, the discriminator should not be able to perform better than chance. The generator is imposed with the loss function given by,

$$\mathcal{L}_{\mathcal{G}}(M,N) = \sum_{k=1}^{Xscales} L_{bce}(D_k(M_k, G_k(M_k), 1)) \quad (3)$$

(3) **Image Gradient Loss :** To account for the sharpness of the image predictions the authors in [12] define a new loss function that accounts for the disparity between the predicted image and the ground truth image by penalizing the differences between the neighboring pixel intensities. This leads to a new loss function given by,

$$\mathcal{L}_{GDL}(\hat{Y}, Y) = \sum_{i,j} ||Y_{i,j} - Yi-1,j| - |\hat{Y}_{i,j} - \hat{Y}_{i-1,j}|| + ||Y_{i,j-1} - Yi,j| - |\hat{Y}_{i,j-1} - \hat{Y}_{i,j}||$$

3.4 Stable Training for GAN

Training generative adversarial nets can be unstable, the network can experience a "mode collapse" where the generator keeps generating a similar data point, and because the data points generated are distinguished by the discriminator with no correlation between them, the discriminator fails to identify whether the data is real or fake, this prompts the generator network to generate same examples. Various techniques have been proposed in [17] to train GAN more efficiently. We adopt certain strategies described in the literature [12, 17] to train GAN in a more stable manner.

(1) Minibatch discrimination - We train the discriminator network in multiple minibatches of size $B = 10$. This allows the discriminator to look at the examples in a collective manner instead of drawing inferences from isolated examples.
(2) Modified Generator Loss function - The generator loss function described in section 3.3 (equation 3) is modified to incorporate the l_p function, thus making the loss function of the generator as follows, $\mathcal{L}adv = \lambda_{adv}\mathcal{L}_{\mathcal{G}} + \lambda_{lp}\mathcal{L}_p$. where λ_{adv} and λ_{lp} are tunable parameters.

4 ANOMALY DETECTION VIA FUTURE FRAME PREDICTION

In this section we discuss in detail the proposed methodology and implementation details. Consider a sequence of frames $I_1, I_2, I_3, ...I_N$ which serve as input, the goal is to predict the future video frames $\hat{I}_{x+1}, \hat{I}_{x+2}, ...\hat{I}_{x+N}$. The network is fed with normal events $I_1, I_2, ...I_N$, this allows the generator to generate frames which contain only normal events. The temporal information is preserved since a sequence of frames are fed as input. We assess the quality of the images generated and use that criteria to categorize whether the frames are anomalous. We use Peak Signal to Noise Ratio (PSNR) as a criteria to measure the quality of the images generated by our network. A higher PSNR means the frame is a normal frame and a lower PSNR value implies the frame is anomalous. The logic behind using PSNR as an assessment criteria is that, since our network is conditioned on only normal events, the network produces high quality frames which have normal activities, thus leading to a higher PSNR value, but a frame containing abnormal activity which was not seen by our network before will lead to generation of frames which have lower PSNR value. The video prediction framework is imposed with a combined loss function from section 3.3 and 3.4 as follows :

$$\mathcal{L}_{combined} = \lambda_{adv}\mathcal{L}_{adv}(M,N) + \lambda_{lp}\mathcal{L}_p(M,N) + \lambda_{gdl}\mathcal{L}_{GDL}(M,N) \quad (4)$$

The PSNR is calculated using the following equation,

$$PSNR(I_i, \hat{I}_i) = 10 \log \frac{(max_{I_i})^2}{1/N \sum_{i=0}^{N}(I_i - \widehat{I_i})} \quad (5)$$

We also introduce Earth Mover's Distance as one of the novel metric to assess the quality of the frames generated, we provide a quantitative evaluation comparing PSNR and EMD.

4.1 Datasets and Experiments

In this section we describe the datasets used to train and test our proposed framework. We use the following datasets :

(1) **UCSD pedestrian dataset 1:** The UCSD pedestrian dataset [11] that contains two parts UCSDped1 containing 34 training videos and 36 testing videos with their pixel-level ground truth masks.
(2) **UCSD pedestrian dataset 2:** The UCSD pedestrain dataset2 which contains 16 training and 12 testing videos with their pixel-level ground truth maps. The UCSD pedestrian dataset consists of various irregular activities like people using bicycles, skate boards and cars on pathways that is meant for pedestrians. We train our model using both the dataset parts.

4.2 Evaluation Metrics

In this section we present the various evaluation metrics used to assess the framework. As discussed in the beginning of section 4, we use PSNR values to assess the quality of the generated frames. Figure 3 shows the plot between PSNR values and the video frames, from the plot we can see that frames which have normal activities (under the realm of training data) have higher PSNR values because our framework has accurately learnt the internal representation of

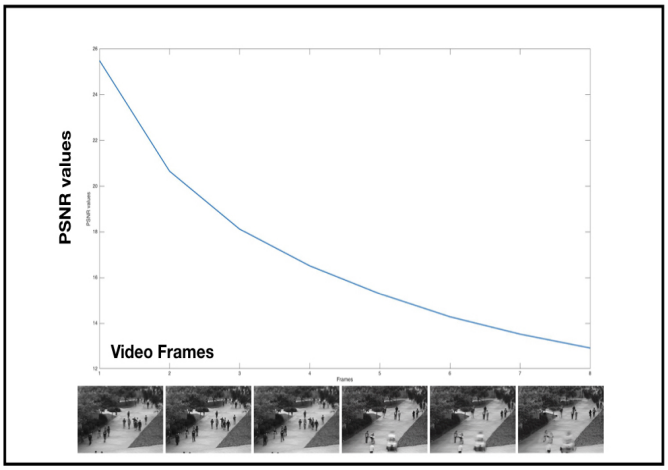

Figure 4: The plot depicts PSNR values vs video frames. The PSNR value is high for frames containing normal activities and decreases for frames that have highly irregular activities.

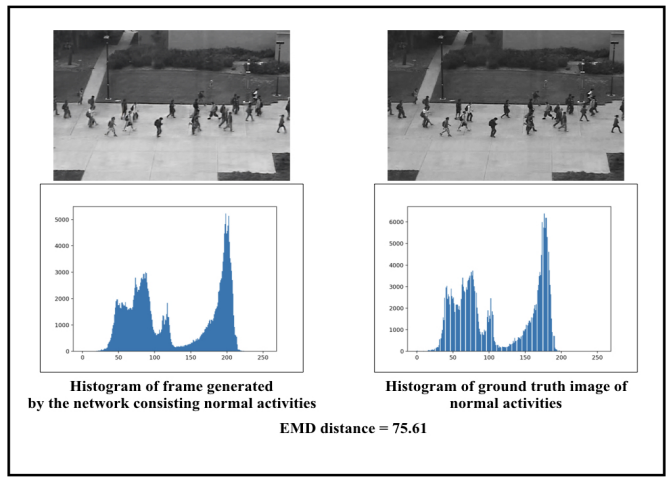

Figure 5: The EMD distance (EMD = 75.61) computed between these two signatures indicate that the generated frame which consists of normal activities is closer to the ground truth.

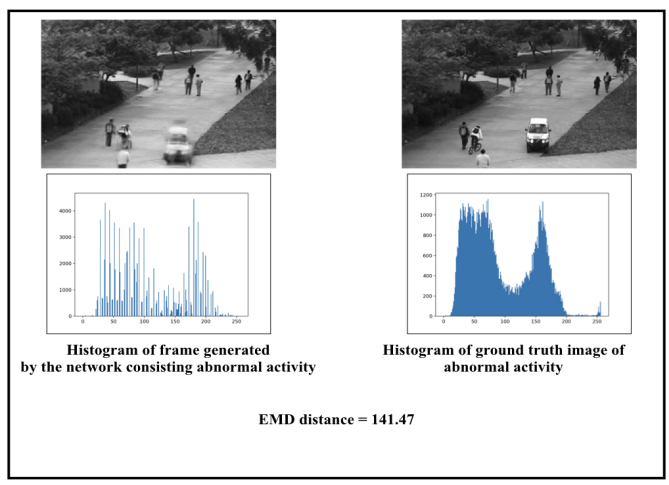

Figure 6: The EMD distance between the generated frame consisting of abnormal events and ground truth reveals a value of 141.47 which is significantly higher to EMD distance of normal frames indicating it is dissimilar and anomalous.

video frames that have normal activities thus successfully generating frames of higher quality while the frames that have irregular activities have lower PSNR values. We also use the evaluation codes provided by the datasets to measure the effectiveness of our framework. The UCSD pedestrian dataset has pixel-level ground truth maps, we compare our generated frames with the ground truth maps to evaluate how our framework is performing.

(1) **Regularity Score** - The regularity score is given by,

$$R(i) = \frac{PSNR(I_i, \widehat{I_i}) - min_i PSNR(I_i, \widehat{I_i})}{max_i PSNR(I_i, \widehat{I_i}) - min_i PSNR(I_i, \widehat{I_i})} \tag{6}$$

(2) **Comparison with ground truth maps** - The frames generated are compared with pixel-level ground truth maps to determine whether the frame is anomalous or not.

(3) **Earth Mover's Distance** - We introduce Earth Mover's Distance as one of the evaluation criteria to assess the frames generated. It is given by,

$$EMD(P, Q) = \frac{\sum_{i=1}^{m} \sum_{j=1}^{n} f_{i,j} d_{i,j}}{\sum_{i=1}^{m} \sum_{j=1}^{n} f_{i,j}} \tag{7}$$

5 RESULTS

In this section we present the results using the evaluation criteria outlined in section 4.2.

5.1 Regularity Score

The PSNR score obtained is normalized to have values between $0 < R < 1$ and is termed as a regularity score, a threshold can be set for the regularity score to distinguish the anomalous and non-anomalous events. It can be calculated as mentioned in the literature [10] as follows

$$R(i) = \frac{PSNR(I_i, \widehat{I_i}) - min_i PSNR(I_i, \widehat{I_i})}{max_i PSNR(I_i, \widehat{I_i}) - min_i PSNR(I_i, \widehat{I_i})} \tag{8}$$

5.2 Comparison with existing methods

In this section we compare our framework with existing handcrafted features methods and state-of-the-art deep learning frameworks. In most of the anomaly detection frameworks proposed [9], the AUROC curve gives a sense of how accurate the framework performs. Since this is a video prediction framework, a direct True Positive Rate and False Positive Rates are tricky to calculate. As a

Table 1: Comparison of AUC values of different methods.

Method	UCSD Ped1	UCSD Ped2
Liu et al. [10]	83.1%	95.4%
Conv-AE	75%	85%
Unmasking	68%	82.2%
Stacked RNN	N/A	92.2%
Hinami et al.	N/A	92.2%
Our Method (PredGAN)	84.27%	93.0%

remedy, we use the regularity scores derived from the PSNR values mentioned in section 4.2. The PSNR values of all the test frames are calculated and consequently the regularity score. The frames generated by the framework consisting of irregular events have lower PSNR values compared to the normal-events' frames. A varying threshold for the regular scores yields the AUC as depicted in the Table1. From the table, we can clearly see that the proposed framework outperforms rest of the methods and is comparable to the work proposed in [10].

5.3 Comparison with ground truth maps

This is yet another evaluation metric that we use to measure the effectiveness of our framework. The datasets used for training and testing provide ground truths that are compared with the generated frames to check whether the prediction is accurate. The UCSD pedestrian dataset provides pixel-level ground truth maps which are compared with the predictions made by the network. The results are presented in section 5.

5.4 Earth Mover's Distance

We introduce Earth Mover's Distance (EMD) as one of the metrics for assessing the quality of the frames generated and also to tag them as anomalous or not-anomalous. EMD is a method to evaluate the dissimilarity between two multi-dimensional distributions. The earth mover's distance intuitively describes the minimum amount of work that needs be done on one distribution to transform into the other distribution under consideration. Recently EMD has found applications in computer vision for tasks like Image retrieval [15, 16] etc. The Earth Mover's Distance is calculated as follows. Consider two distributions P and Q each with m and n clusters respectively,

$P = \{(p_1, w_{p1}) ...(p_m, w_{pm})\}$ and $Q = \{(q_1, w_{q1}), ...(q_n, w_{qn})\}$, where p_1, w_{p1}, q_1, w_{q1} represent the cluster index and weight of the cluster for both the distributions respectively. Let $d(i, j)$ represent the ground distance between both of the distributions. Now the objective is to find a flow F between p_i and q_j that minimizes the overall effort for the conversion. It can be mathematically represented as follows,

$$Convert(P, Q, F) = \sum_{i=1}^{m} \sum_{j=1}^{n} f_{i,j} d_{i,j} \tag{9}$$

The earth mover's distance is then given by,

$$EMD(P, Q) = \frac{\sum_{i=1}^{m} \sum_{j=1}^{n} f_{i,j} d_{i,j}}{\sum_{i=1}^{m} \sum_{j=1}^{n} f_{i,j}} \tag{10}$$

We use this distance to measure the dissimilarity between the ground truth frames and the generated video frames. A larger EMD value implies the generated frame is more dissimilar to the ground truth frame. The frames which are generated when the network is conditioned on normal events will have a lower EMD value when compared with their ground truths while frames which have anomalous events will result in higher EMD value when compared to their corresponding ground truth as the network fails to predict the future frames of the irregular activities. We calculate the histograms of the ground truth frame and the generated frame and use this histogram as a signature to compute the Earth Mover's Distance.

In figure 4 and 5 we have plotted a histogram for the predicted frames and their corresponding ground truths. The EMD distance is calculated for both the signatures and we see that in figure 4 which contains normal activity, the EMD score is lower (75.61) indicating there is a similarity between the ground truth while in figure 5 we see the EMD value is 141.47, as expected for frames consisting of abnormal activity. In figure 6 and 7, we have compared the predicted frames with their corresponding pixel-level ground truth maps supplied by the dataset, and its evident that our framework has been successful in identifying even subtle abnormalities like cycle riders and wheel chair with remarkable accuracy. In figure 8 we have compared the predicted frame with the anomaly localization code provided by the avenue dataset, we can see that our predictions match the ground truth.

5.5 EMD vs PSNR

In this section, we explain why Earth Mover's Distance is a better evaluation metric to assess the quality of the images generated by the network. The Peak Signal to Noise Ratio which is primarily calculated using the Mean Square Error [4] is a quick way to assess the quality of the images, but the problem is that they aren't agreeable with the way the Human Vision System (HVS) perceives the distortion in images. This is demonstrated by calculating the PSNR value of a frame generated by the network and is subjected to a gamma correction, consequently shifting the image from the darker end of the spectrum to lighter spectrum. From Figure 7, it is clear that though the perceptual similarity of the image deteriorates the change in the PSNR value is minimal, on the contrary, the EMD fluctuates rapidly indicating that it is in complete agreement with the HVS. This ability of the EMD to capture the perceptual similarity of the images makes it a more suitable candidate to assess the quality of the images generated by the prediction framework.

6 CONCLUSION

In this paper we have proposed a video prediction framework called PredGAN which is trained adversarially to detect anomalous activities in videos. We hypothesize that a video prediction framework which is conditioned to learn normal activities can be used to accurately predict the evolution of those frames and when the network encounters video frames with irregular activities the frames predicted are fuzzy. We have also introduced various evaluation metrics like PSNR, Comparison with ground truth maps and Earth Mover's Distance (EMD) to assess the performance of the framework. We have also shown how EMD is a better metric to assess the quality of the images generated by the prediction network compared to

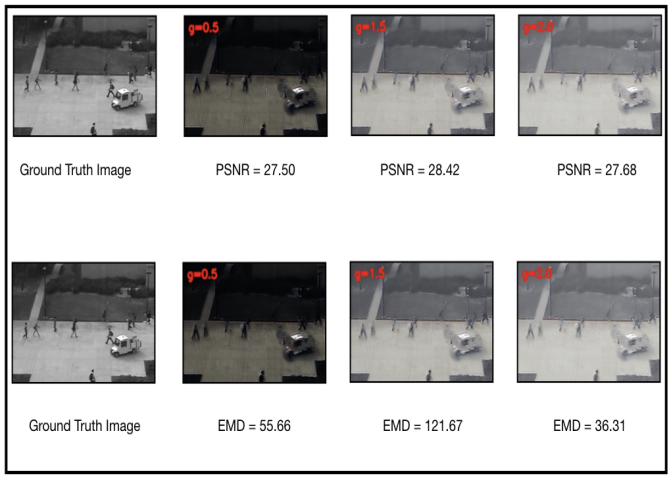

Figure 7: A comparison between the PSNR and EMD yields the fact that EMD better captures the perceptual similarity of the images and is much closer to the Human Vision System making it apt for assessing the quality of the images generated by the prediction framework.

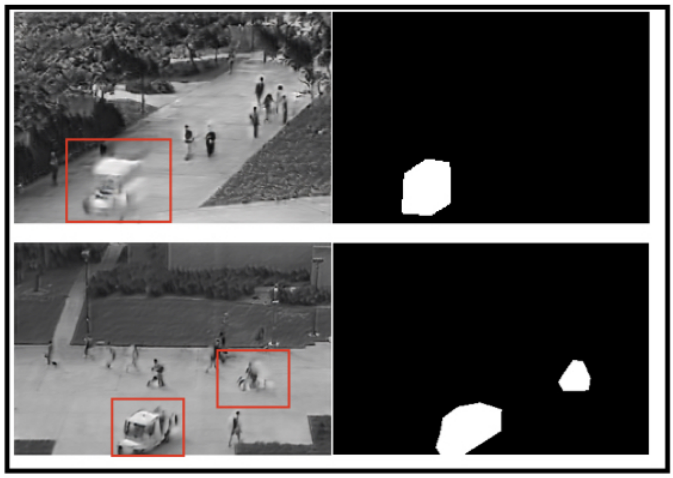

Figure 8: A comparison between the pixel-level ground truth maps and the frames predicted by the network indicates that our framework is capable of detecting subtle anomalous events like cycles.

PSNR. Extensive experiments carried out on the dataset indicates the effectiveness of our framework.

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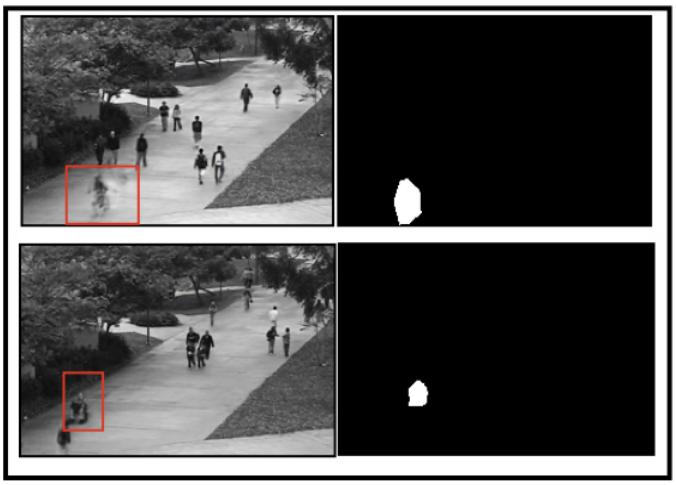

Figure 9: A comparison between the pixel-level ground truth maps and the frames predicted by the network indicates that our framework is capable of detecting subtle anomalous events like wheel chairs.

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